A Rigorous and Stable Implementation of the Crank–Nicolson Method for the TDSE

1. Physical Problem: Time Evolution in Quantum Mechanics

The time-dependent Schrödinger equation (TDSE) governs the evolution of quantum states:

$$i\hbar\frac{\partial\Psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\Psi(x,t) \tag{1}$$

Key Requirement: The total probability must be conserved:

$$\int |\Psi(x,t)|^2 dx = 1$$

Interpretation:

- The Hamiltonian $\hat{H}$ controls energy + dynamics
- Evolution must be **unitary**
- Numerical methods must preserve norm (very important!)

2. Discretization of Space and Time

We discretize:

$$x_j = j\Delta x, \quad t_n = n\Delta t$$

Second derivative (central difference):

$$\frac{\partial^2\Psi}{\partial x^2} \approx \frac{\Psi_{j+1}^n - 2\Psi_j^n + \Psi_{j-1}^n}{(\Delta x)^2} \tag{2}$$

Result: PDE $\to$ System of algebraic equations

Insight:

- Smaller $\Delta x \to$ higher spatial accuracy
- Smaller $\Delta t \to$ better temporal resolution
- Tradeoff: accuracy vs computational cost

3. Why Standard Methods Fail

FTCS (Explicit Method):

- Easy to implement
- But **unconditionally unstable**
- Amplifies numerical noise $\to$ blows up

BTCS (Implicit Method):

- Stable
- But introduces **artificial damping**
- Violates quantum probability conservation

Conclusion: We need a method that is:

- Stable
- Accurate
- Norm-preserving

4. Crank–Nicolson Scheme

The Crank–Nicolson method averages FTCS and BTCS:

$$i\hbar\frac{\Psi_j^{n+1} - \Psi_j^n}{\Delta t} = \frac{1}{2}\hat{H}_{FD}\Psi_j^{n+1} + \frac{1}{2}\hat{H}_{FD}\Psi_j^n \tag{3}$$

Rewriting:

$$\left(I + \frac{i\Delta t}{2\hbar}H\right)\Psi^{n+1} = \left(I - \frac{i\Delta t}{2\hbar}H\right)\Psi^n \tag{4}$$

Key Idea:

- Implicit + symmetric in time
- Second-order accurate in both space and time

5. Unitarity and Stability

Define propagator:

$$U_{CN} = (I + i\gamma H)^{-1}(I - i\gamma H)$$

This is a **Cayley transform** of a Hermitian operator.

Result:

$$U^\dagger U = I$$

Implications:

- Exact probability conservation
- No artificial damping
- No instability

This is why CN is the gold standard for TDSE.

6. Matrix Formulation

Let:

$$\alpha = \frac{i\Delta t}{2\hbar}, \quad \beta = \frac{\hbar^2}{2m(\Delta x)^2}$$

Hamiltonian becomes:

$$H\Psi_j = -\beta\Psi_{j-1} + (2\beta + V_j)\Psi_j - \beta\Psi_{j+1}$$

We obtain:

$$A\Psi^{n+1} = B\Psi^n$$

Important:

- A, B are tridiagonal
- Only nearest-neighbour coupling

7. Efficient Solution: Thomas Algorithm

Instead of inverting matrices:

$$\mathcal{O}(N^3) \to \mathcal{O}(N)$$

Thomas Algorithm Steps:

1. Forward elimination
2. Modify coefficients
3. Back substitution

Why it works:

- Exploits tridiagonal structure
- No unnecessary computations

8. Boundary Conditions

Dirichlet (Infinite Well):

$$\Psi_0 = \Psi_N = 0$$

Periodic:

$$\Psi_0 = \Psi_{N-1}$$

Physical Meaning:

- Dirichlet $\to$ particle confined
- Periodic $\to$ lattice / solid-state physics

9. Algorithm Summary (Implementation)

Step-by-step:

1. Initialize $\Psi(x,0)$
2. Construct matrices A, B
3. For each time step:
   - Compute RHS: $B\Psi^n$
   - Solve $A\Psi^{n+1} = RHS$
   - Normalize (optional check)
4. Repeat

Output: Time evolution of wavefunction

10. Applications

- Quantum tunneling
- Wave packet propagation
- Quantum wells and barriers

Takeaway: Crank–Nicolson provides a perfect balance of:

Accuracy + Stability + Physical Correctness